ML4350 RL Hockey Project

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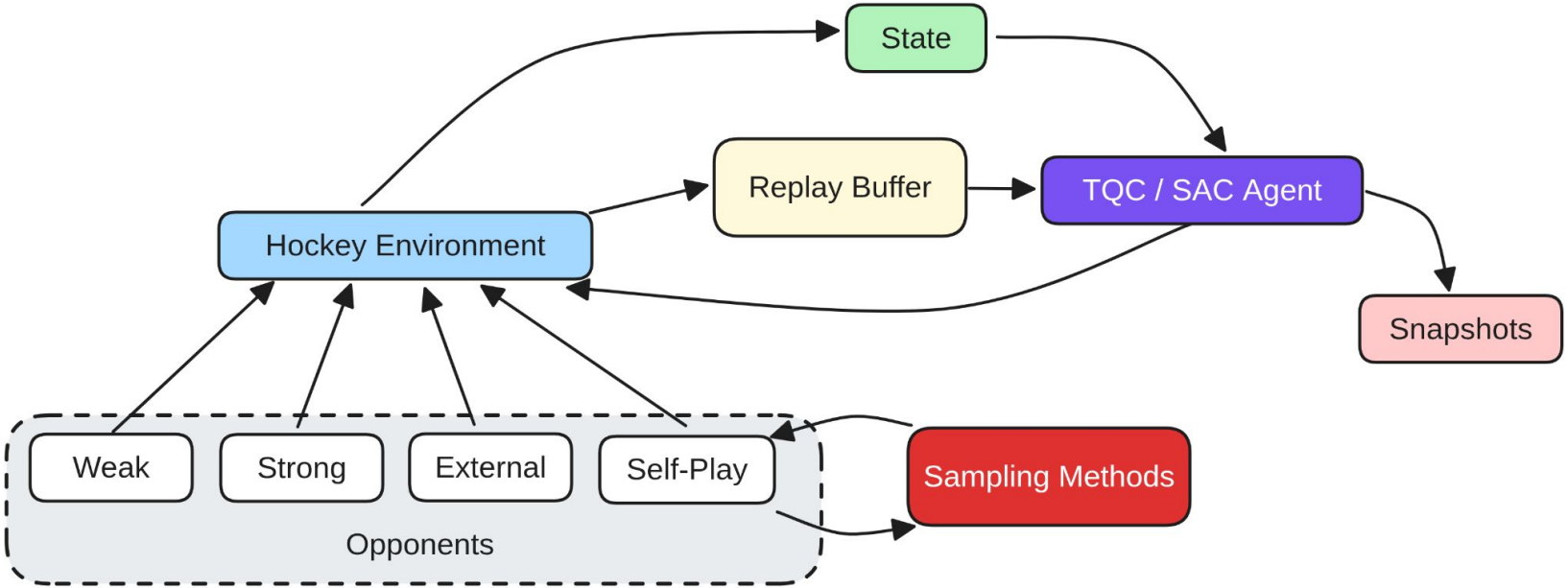
Winter Semester 2025-2026
By Prof. Dr. Georg Martius

Overview

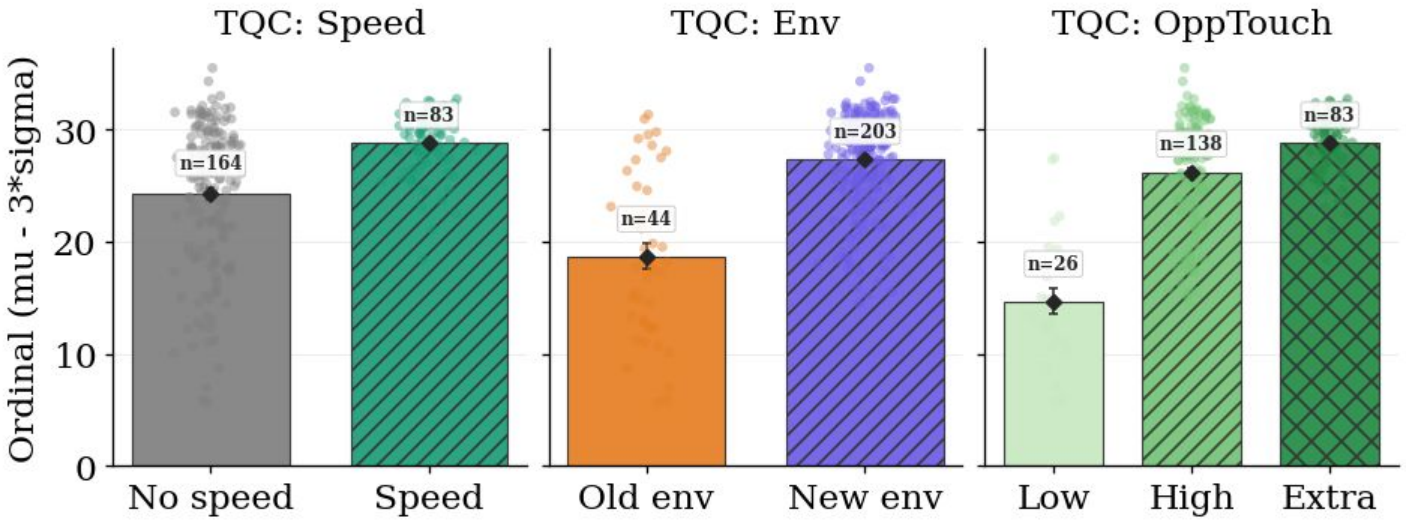


Repository: https://github.com/KouroshKSH/ML4350-RL-Hockey-Agent

Abtin Mogharabin’s Contributions

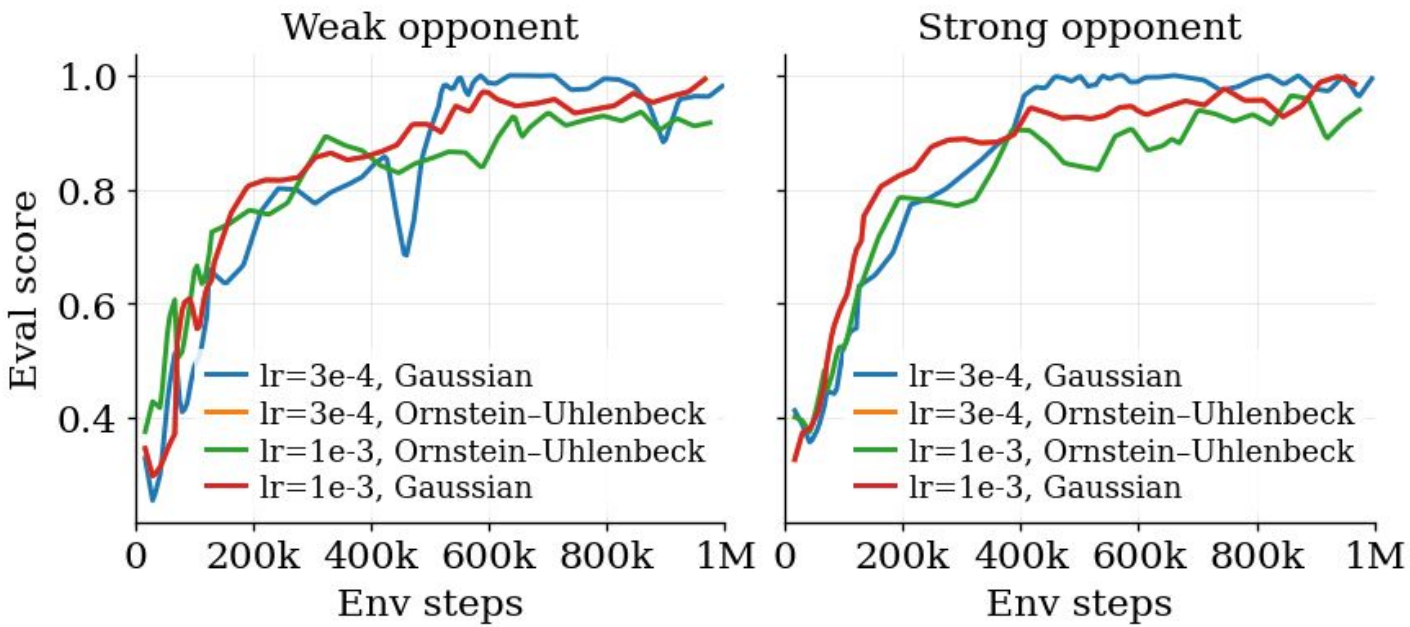
- **Reward Shaping:** (1) Move faster towards an attacking puck, (2) avoid losing the puck to the opponent
- **Environment Modification:** Randomized opponent and puck’s spawning position.
- **Prioritized Fictitious Self-Play (PFSP):** Gives higher probability to self-play opponents where the agent's win rate is around high, to prevent unwanted opponents being ignored.
- **Upper Confidence Bound (UCB):** Prefer opponents that are uncertain and potentially challenging.

Environment & Reward Modification on TQC



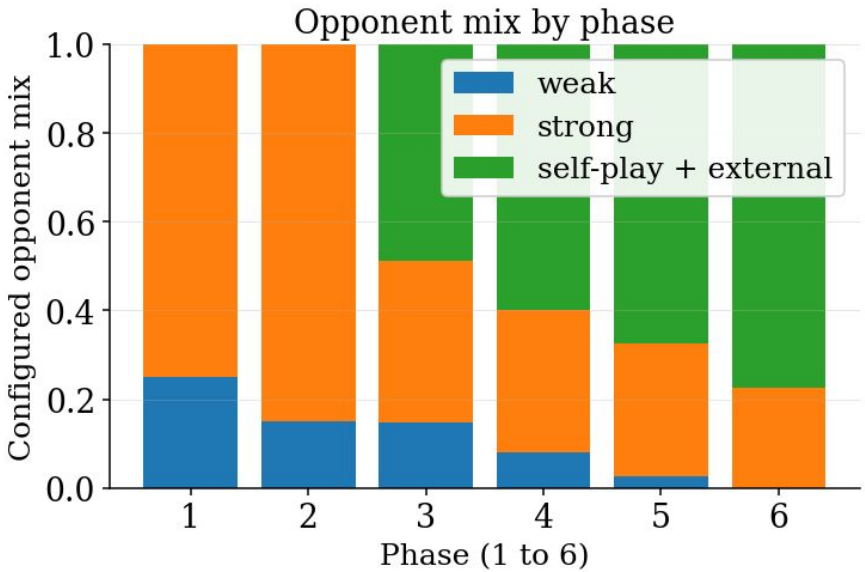
TQC algorithm by Abtin

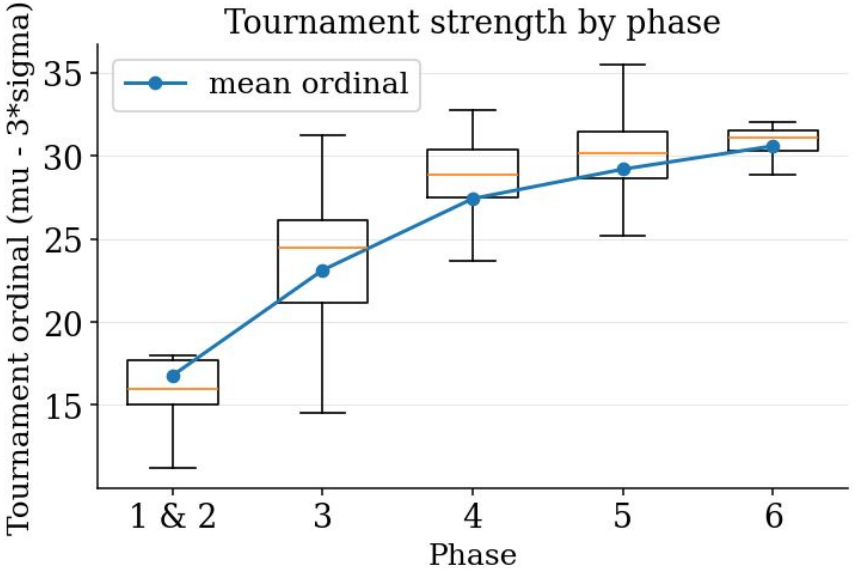
TQC Grid Search on Noise & Learning Rate



TQC algorithm by Abtin

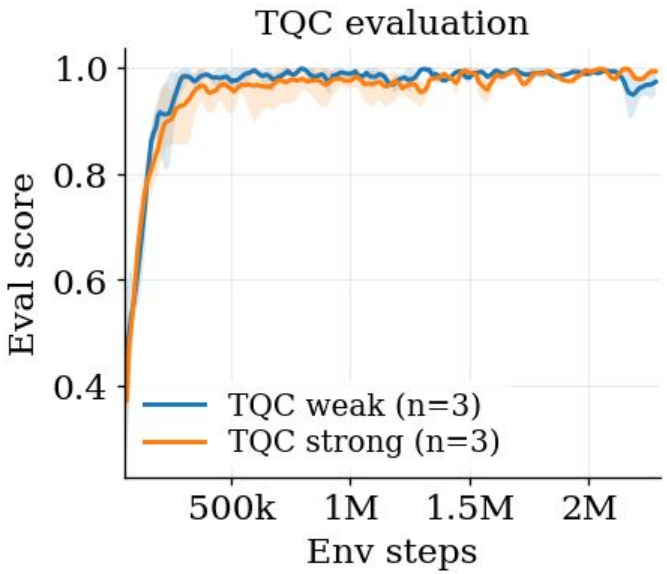
TQC Self Play

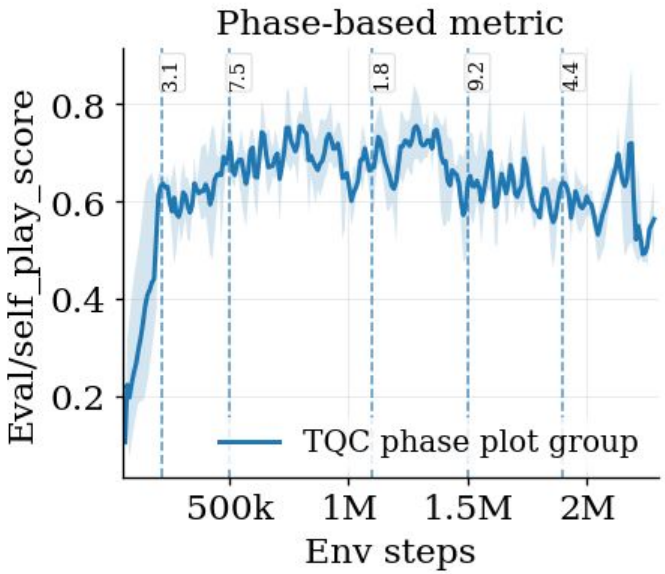




TQC algorithm by Abtin

Training of TQC Agent



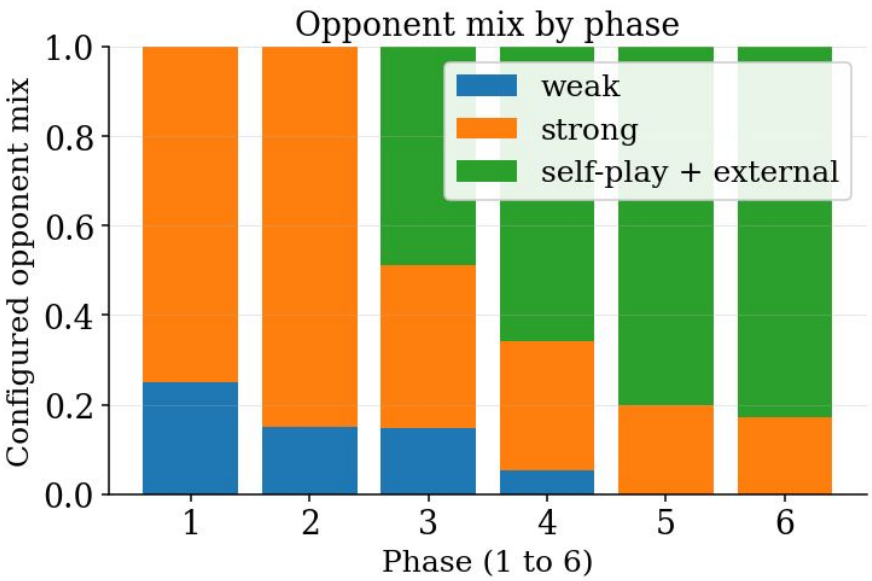


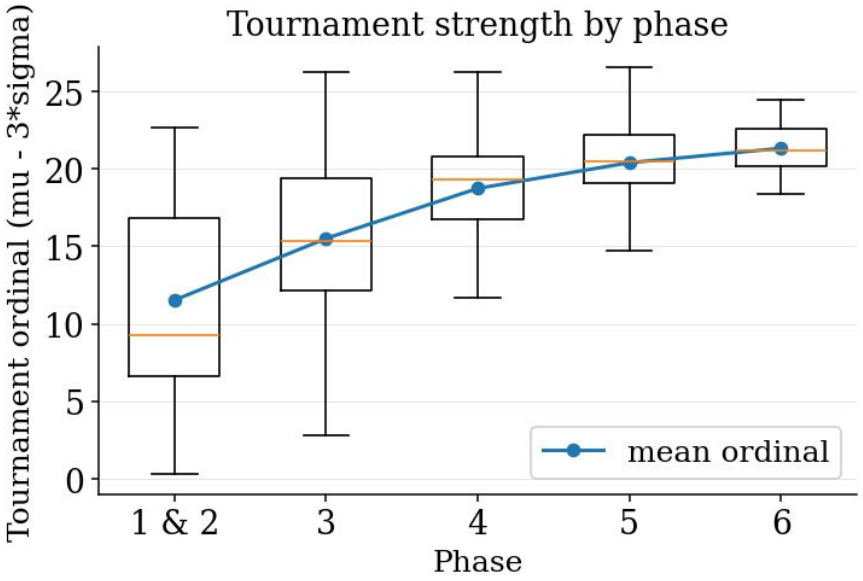
TQC algorithm by Abtin

Kourosh Sharifi's Contributions

- **Progress & Regularization:** (1) Potential-based puck-to-goal progress, (2) L_2 action penalty to ensure smooth movements.
- **Pink Noise Exploration:** Replaces standard Gaussian noise with temporally correlated noise to help the agent discover multi-step maneuvers.
- **PSRO League Training:** Maintains a snapshot pool and uses Replicator Dynamics to solve for a Nash Equilibrium meta-strategy, preventing strategy cycling.

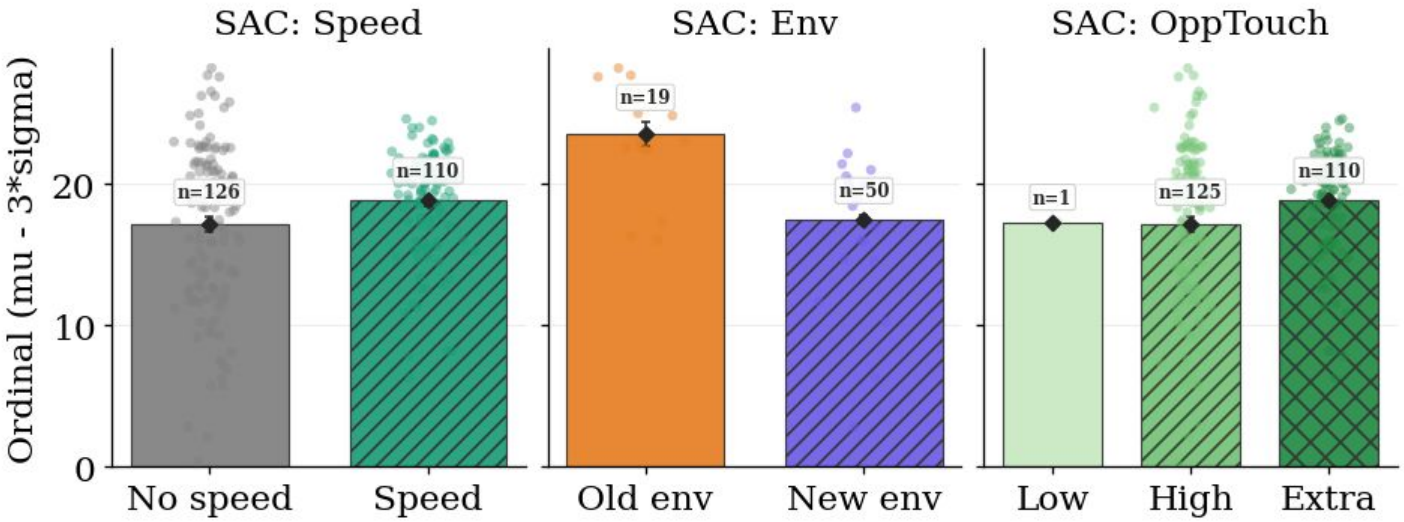
SAC Opponent Pool per Phase





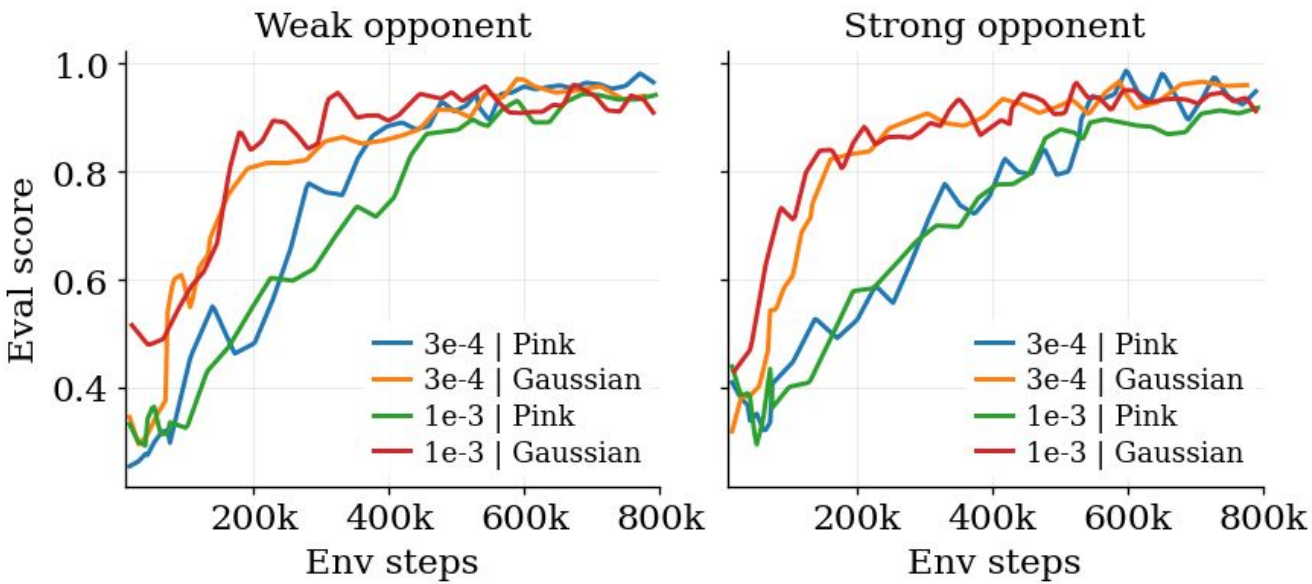
SAC algorithm by Kourosh

SAC Environment & Reward Modification



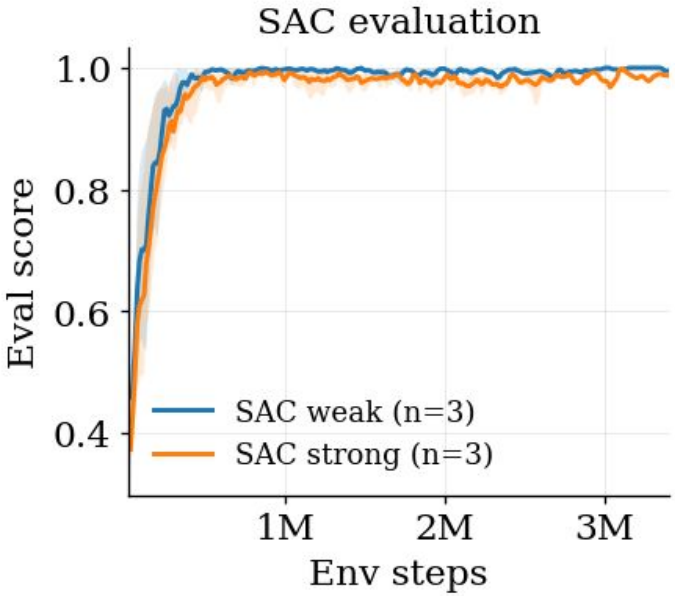
SAC algorithm by Kourosh

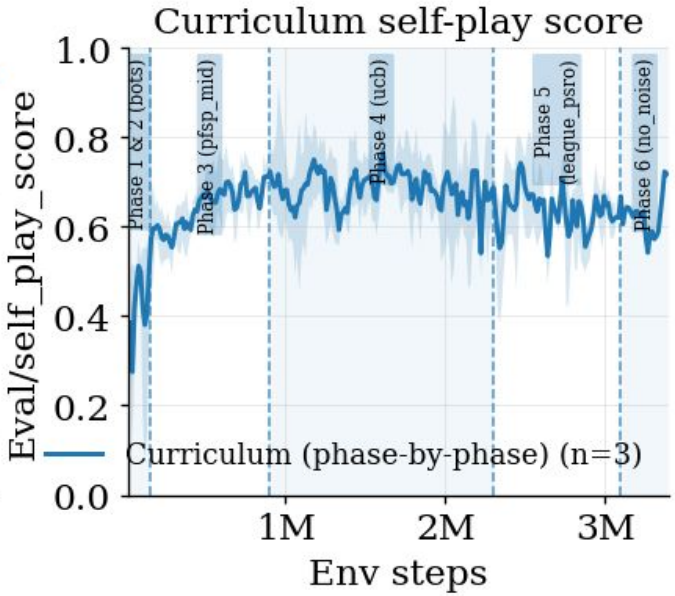
SAC Noise Comparison



SAC algorithm by Kourosh

Evaluation of SAC Agent





SAC algorithm by Kourosh

Thank you for your attention!